# Brainwave Architecture

# Optimization: Persistency

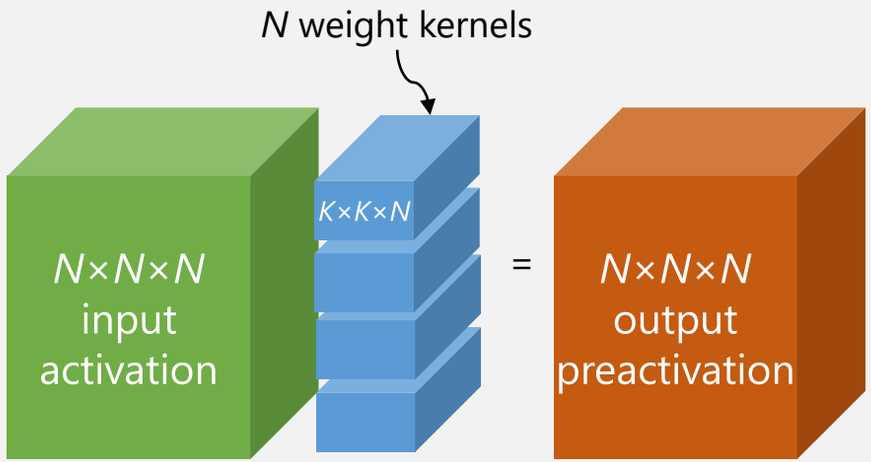
Sparsh Mittal

Images are courtesy of: http://learningsys.org/nips17/assets/slides/brainwave-nips17.pdf

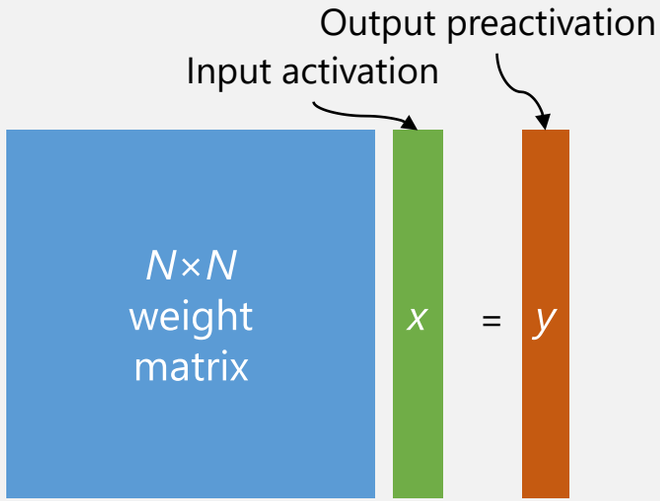
# Acronyms

- MLP – Multilayer perceptron
- LSTM – Long-short term memory
- GRU – Gated recurrent unit
- FPGA – Field-programmable gate array
- DRAM – Dynamic random-access memory
- HW – Hardware
- NPU – Neural processing unit
- DNN – Deep neural network

# Arithmetic Intensity (Compute-to-Data Ratio) of Different Applications




$O(N^3)$ data
$O(N^4K^2)$ compute

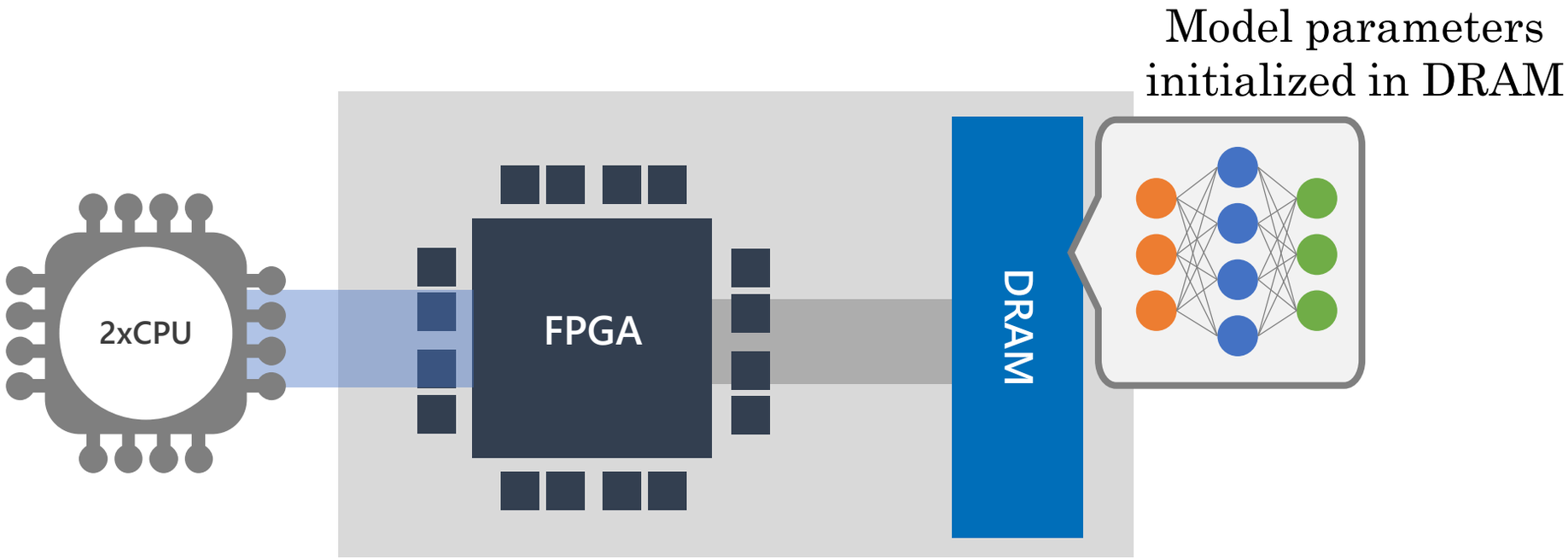



$O(N^2)$ data
$O(N^2)$ compute

Convolutional neural network (CNN)
High compute-to-data ratio

MLPs, LSTMs, and GRUs
Low compute-to-data ratio

# Conventional Acceleration Approach: Local Offload and Streaming



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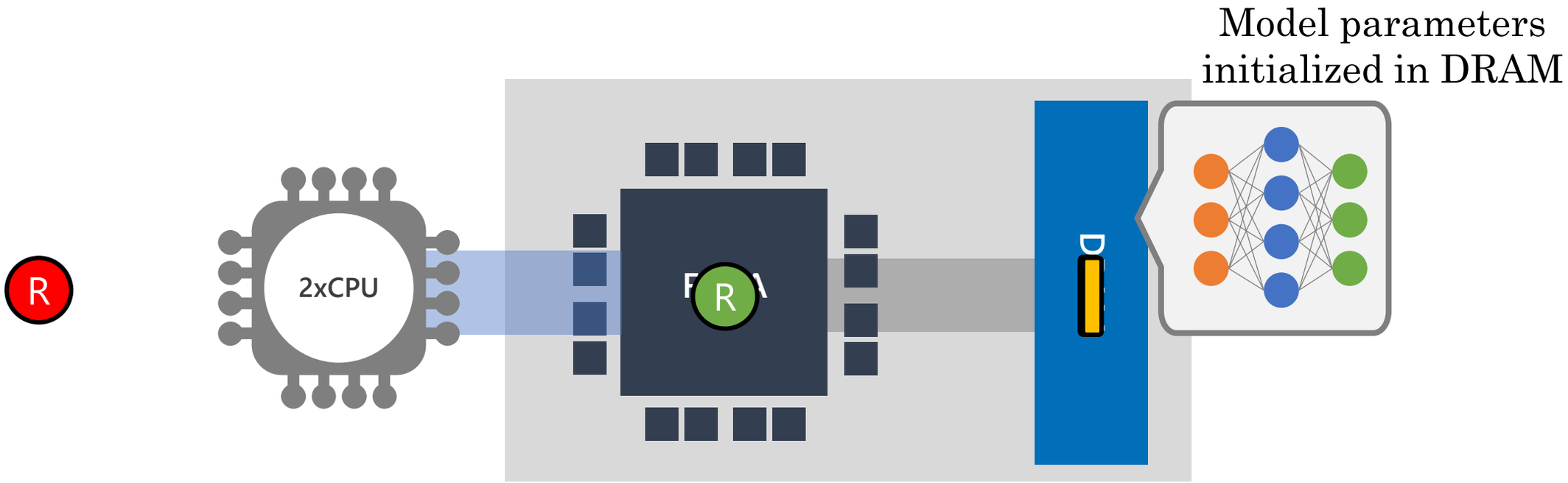
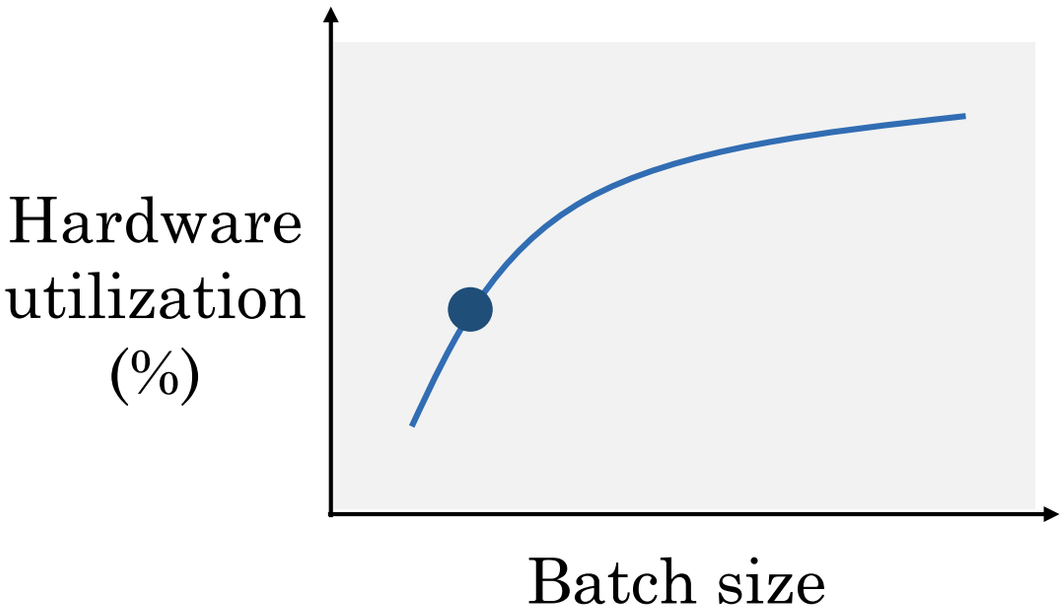
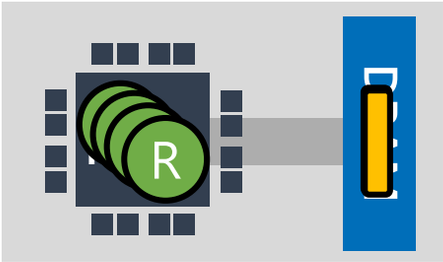



For memory-intensive DNNs with low compute-to-data ratios (e.g., LSTM), hardware utilization is limited by off-chip DRAM bandwidth

# Improving HW Utilization with Batching





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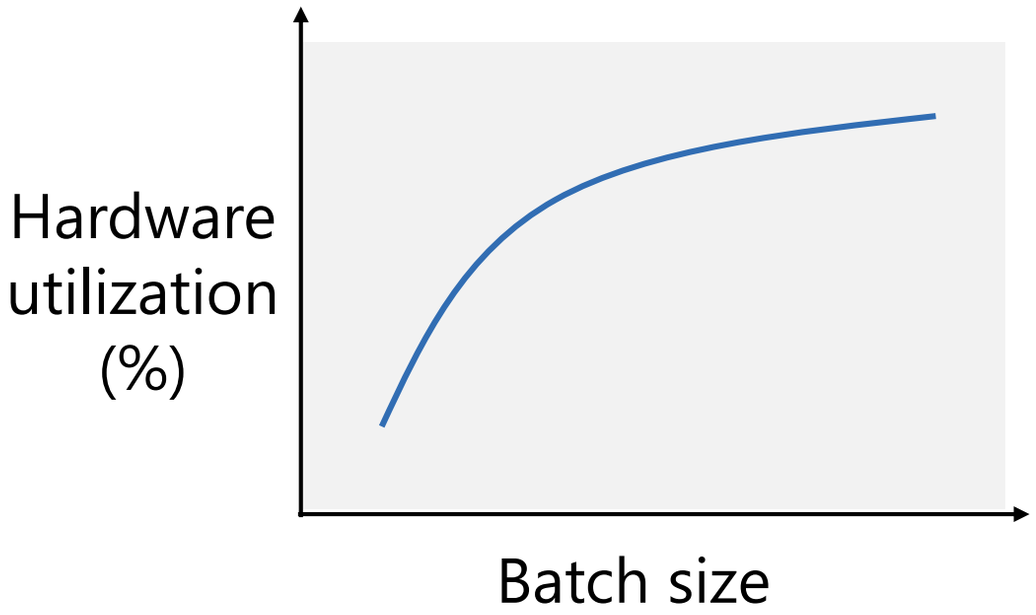


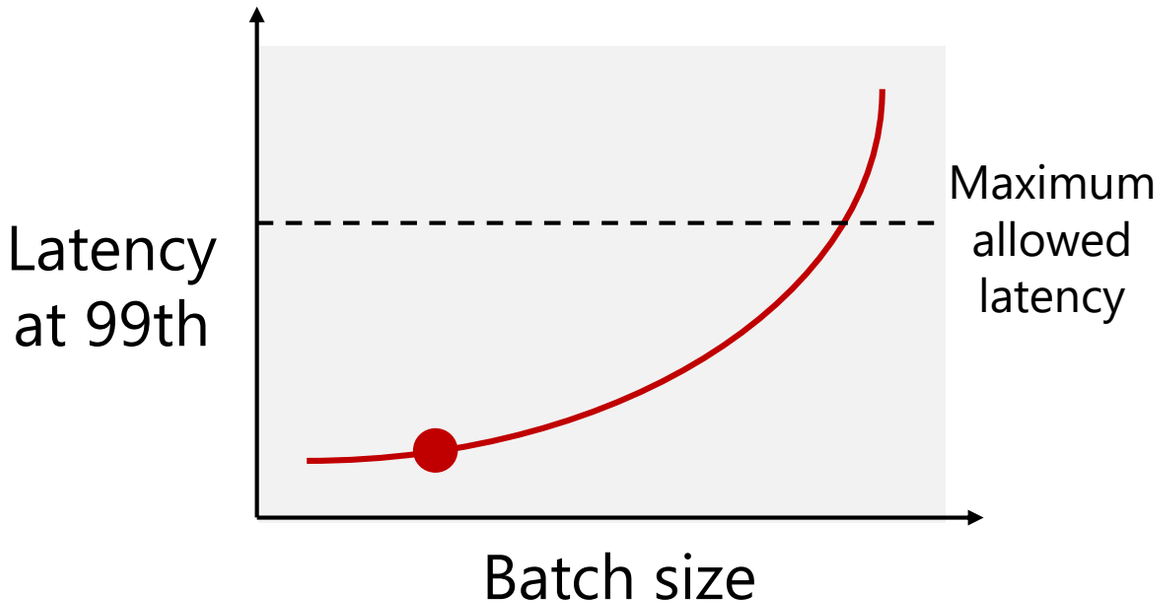


Batching improves HW utilization but increases latency

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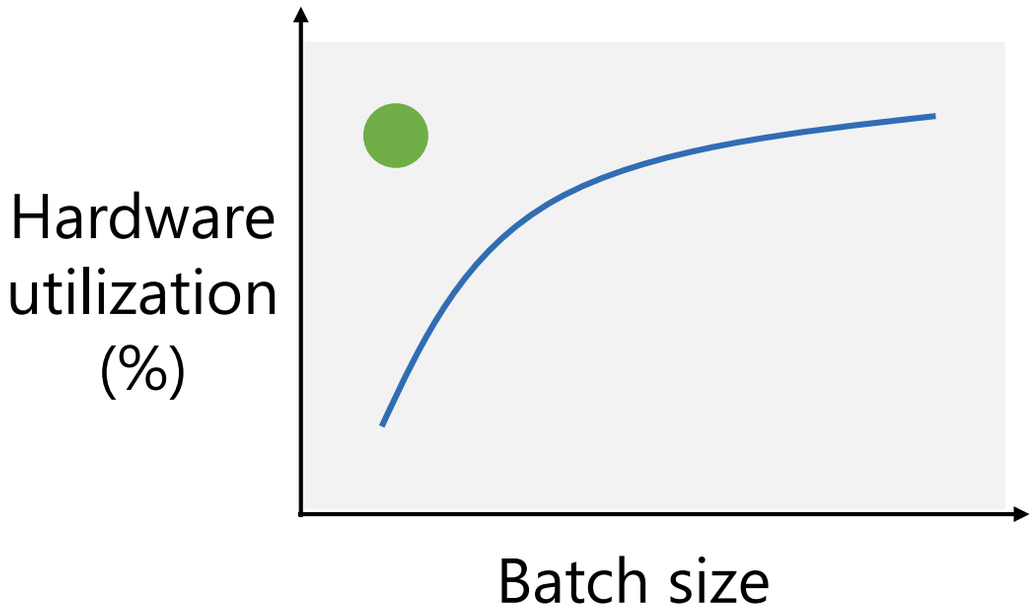
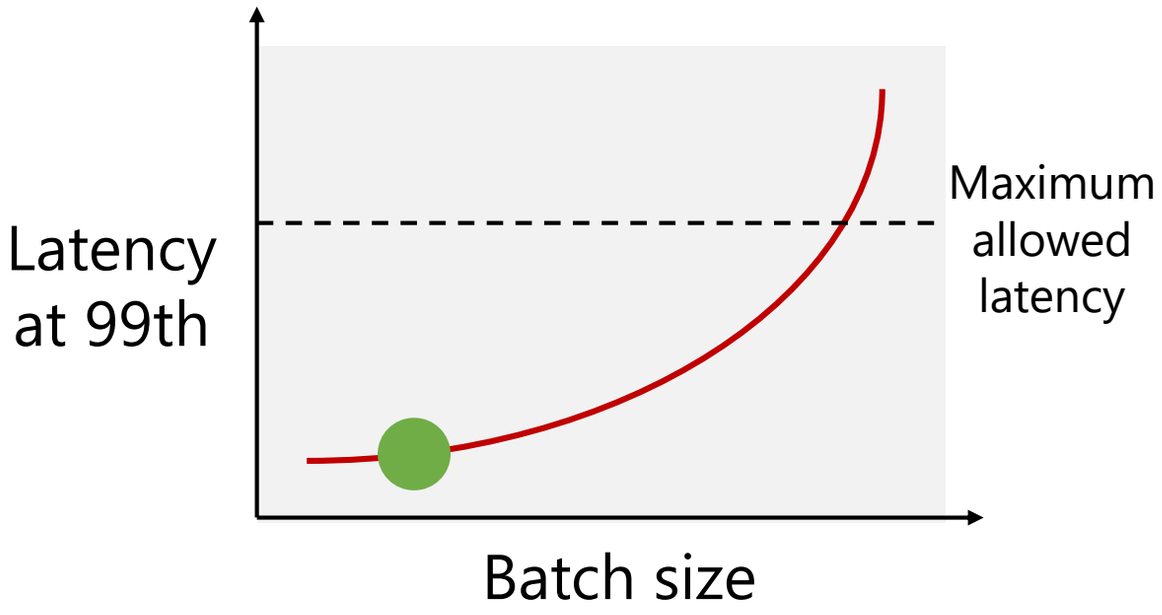
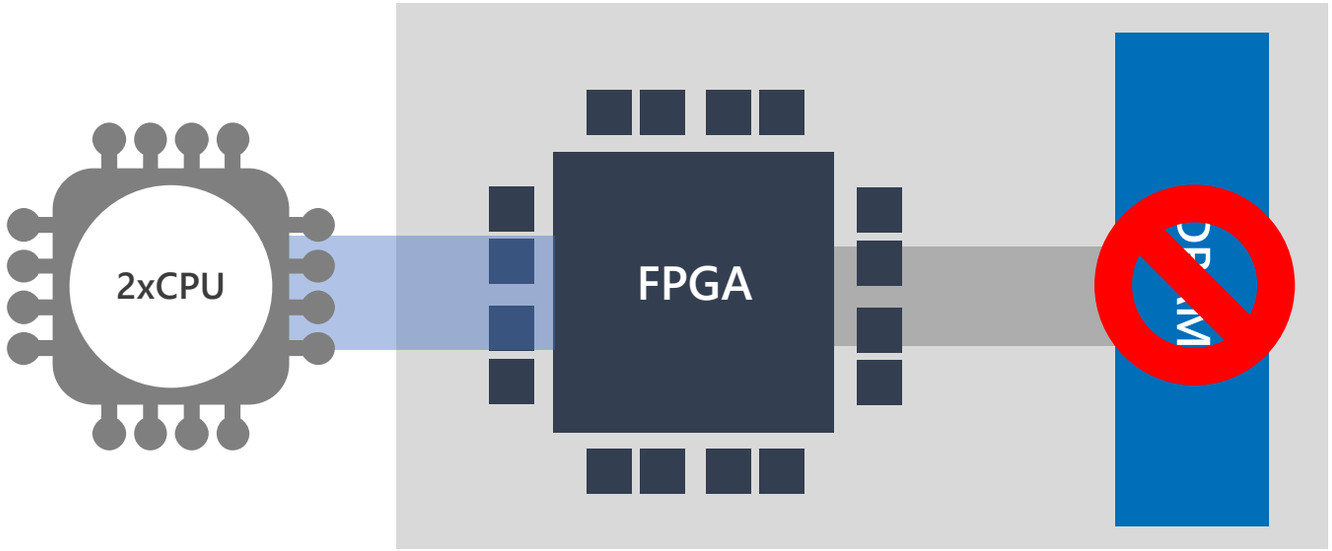



Batching improves HW utilization but increases latency

Ideally, we want high HW utilization at low batch sizes

# Alternative: “Persistent” Neural Nets



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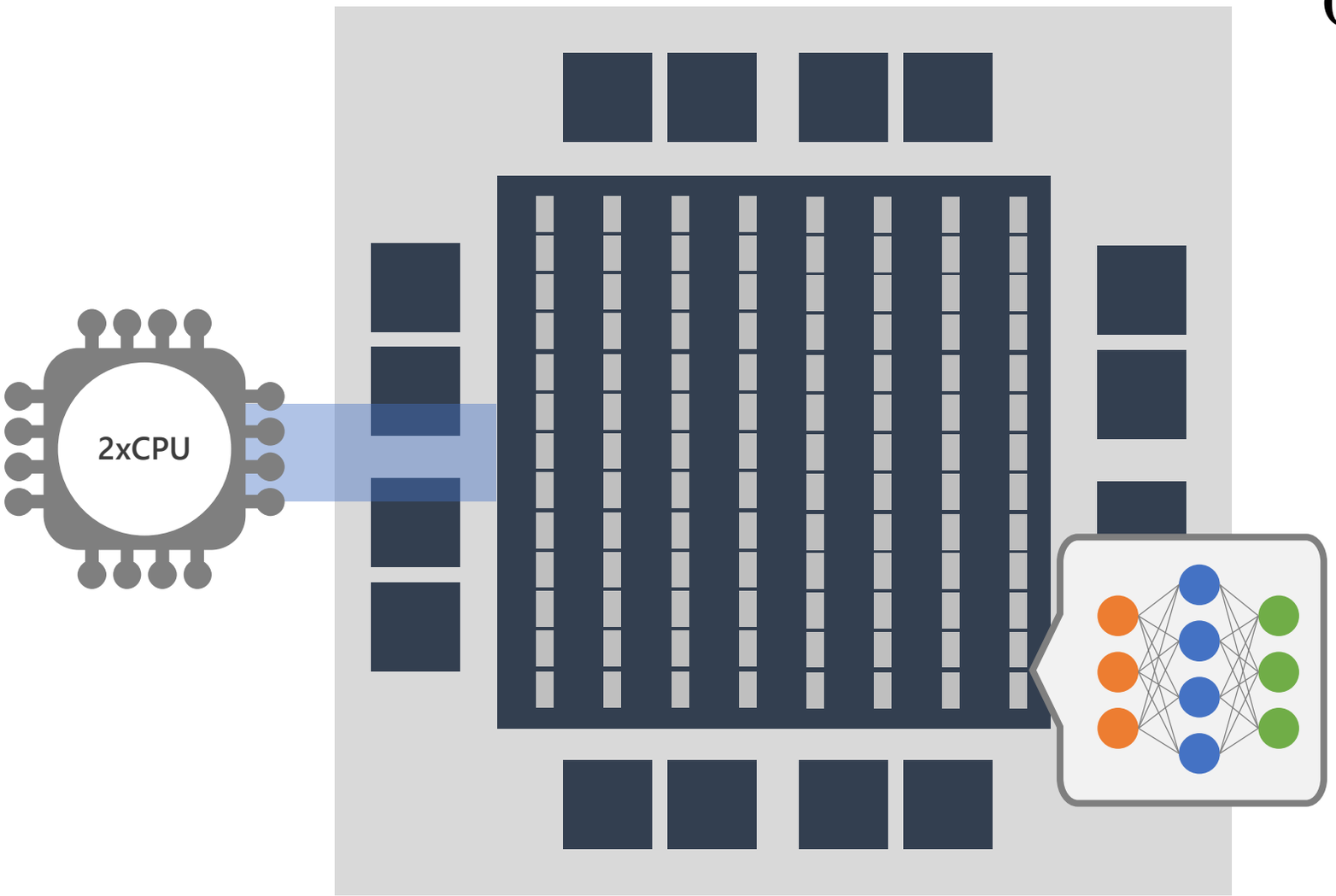


## Observations

State-of-art FPGAs have O(10K) distributed block RAMs, of size O(10MB)
⇒Tens of TB/s of memory BW

Large-scale cloud services and DNN models run persistently

Solution: persist all model parameters in FPGA on-chip memory during service lifetime

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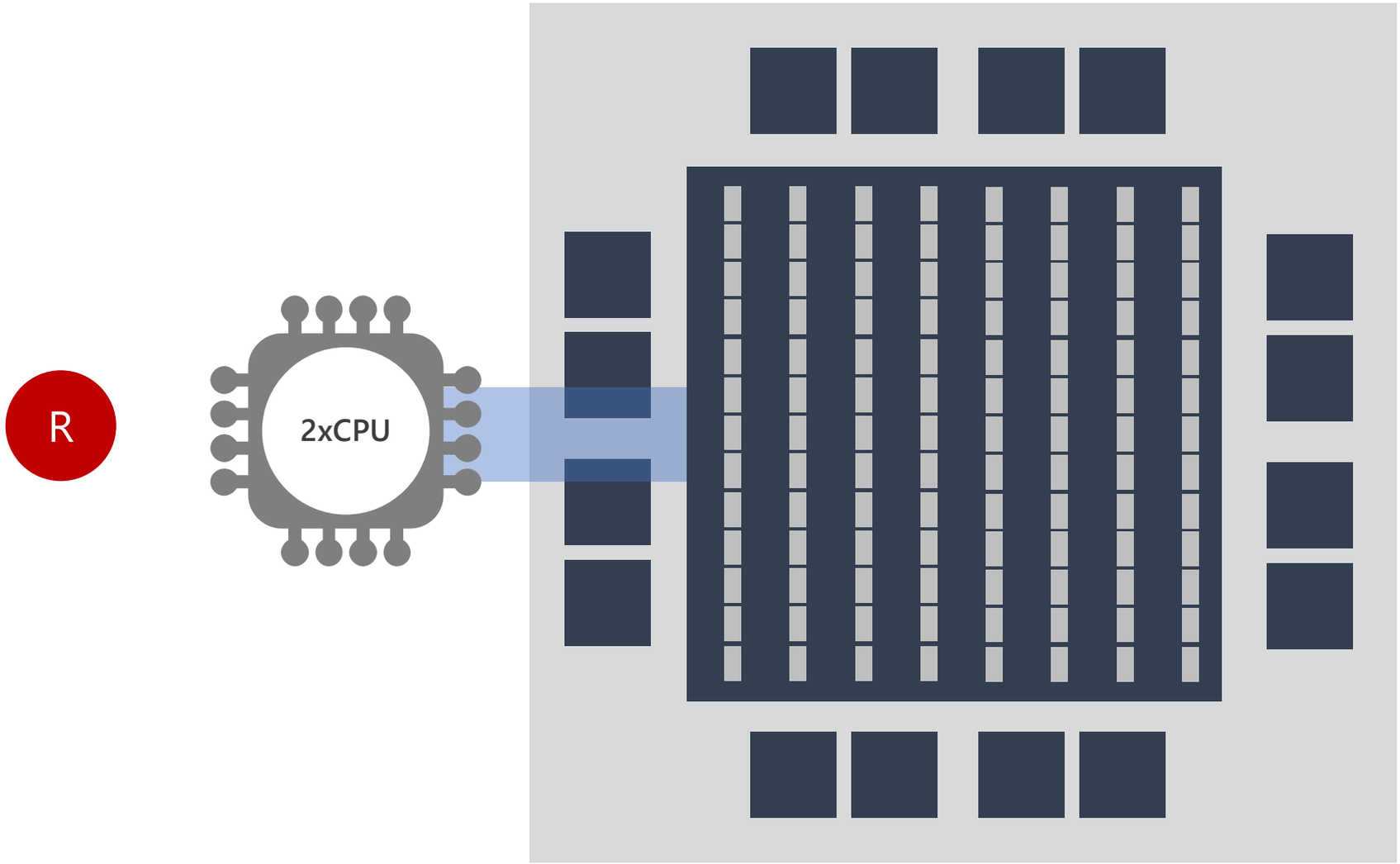

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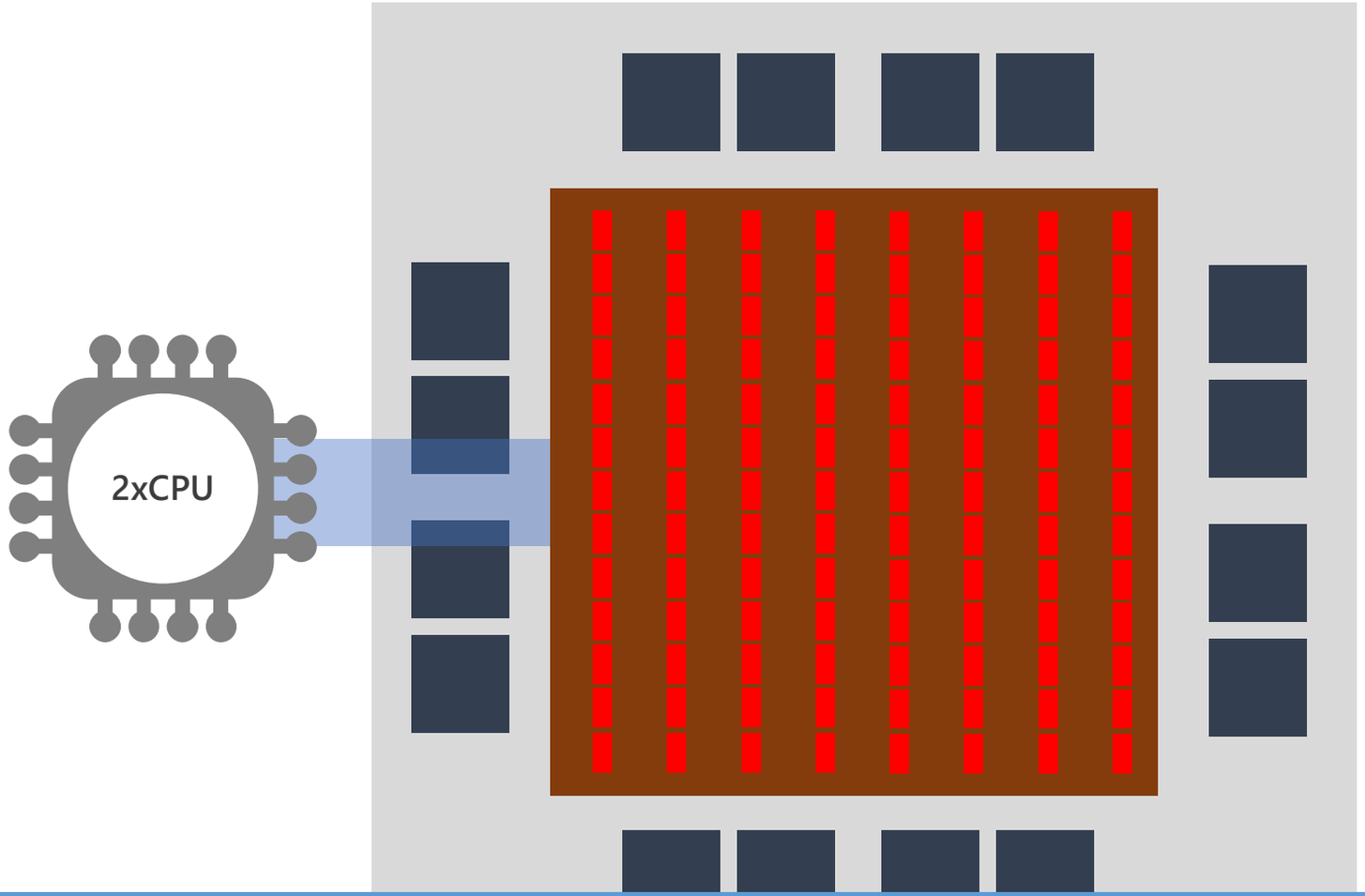



When single request arrives, all chip resources (on-chip memories and compute units) are used to process a single query (no batching required)

# Example

- Intel Stratix 10 280 FPGA has 11,721 independently addressable 512×40-b SRAMs.

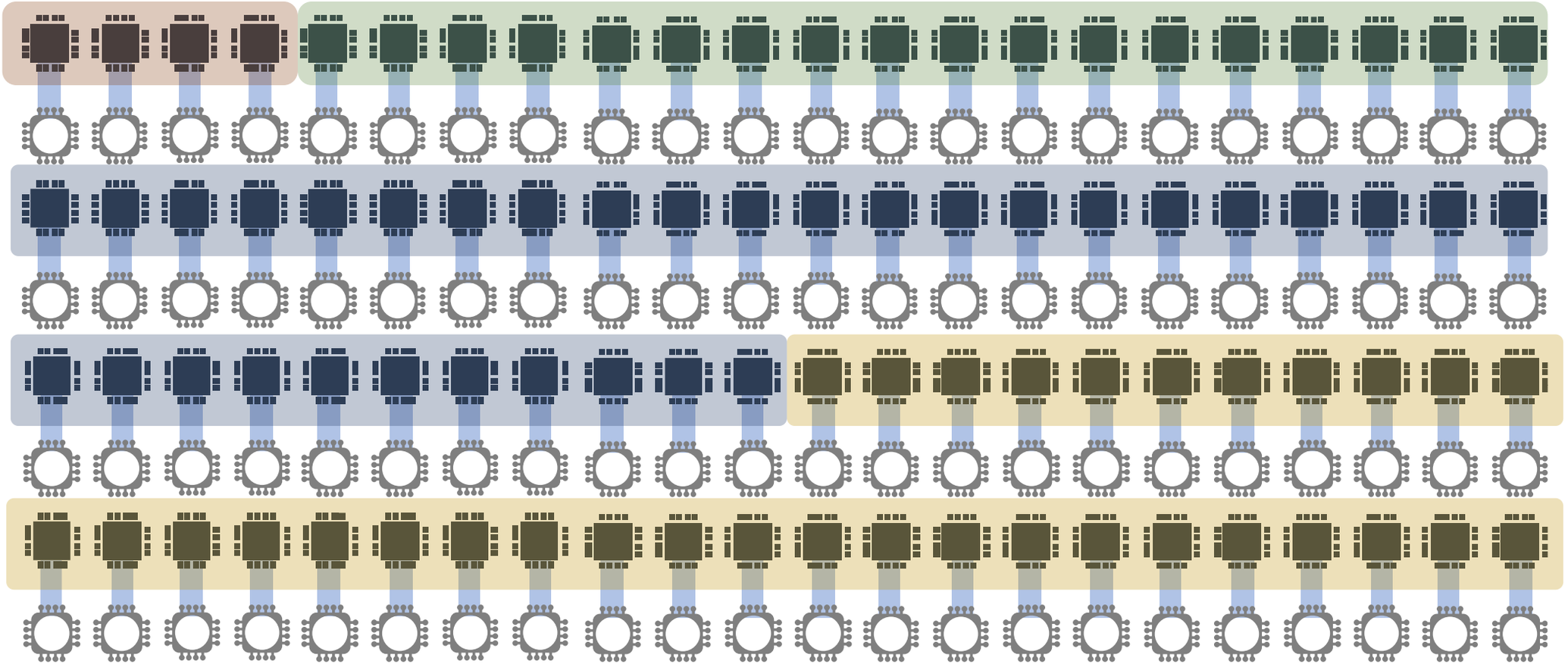
⇒Total 30 MB on-chip storage and 35 TB/s of bandwidth at 600 MHz.

- Typical instances of Brainwave NPUs on Stratix 10 allocate storage for tens of millions of parameters and hundreds of thousands of vector elements.
- While pinning is the default strategy for most NNs, DRAM can also be used for buffering activations and weights.

# What If Model Does Not Fit in Single FPGA?

Solution: Implement persistency at data-center scale

# Solution: Persistency at Data Center Scale



Multiple FPGAs at data center scale can form a persistent DNN HW microservice, enabling scale out of models at ultralow latencies